DS & AI ENGINEERING



Algorithm

Miscellaneous

Lecture No. 03



By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Set
- Union
- Find

Kruskal

Topics to be Covered



- Topic
- Topic
- Topic

Optimal Cost Binary Search Tree

(OBST)

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic: Miscellaneous

Binary Search Tree : (BST)

BST is a Binary Tree, which has the property that the value at every node is less than the value of the right child (and all the values in the right sub-tree) and the value at every node is greater than the value of the left child (and all the values in the left sub-tree).

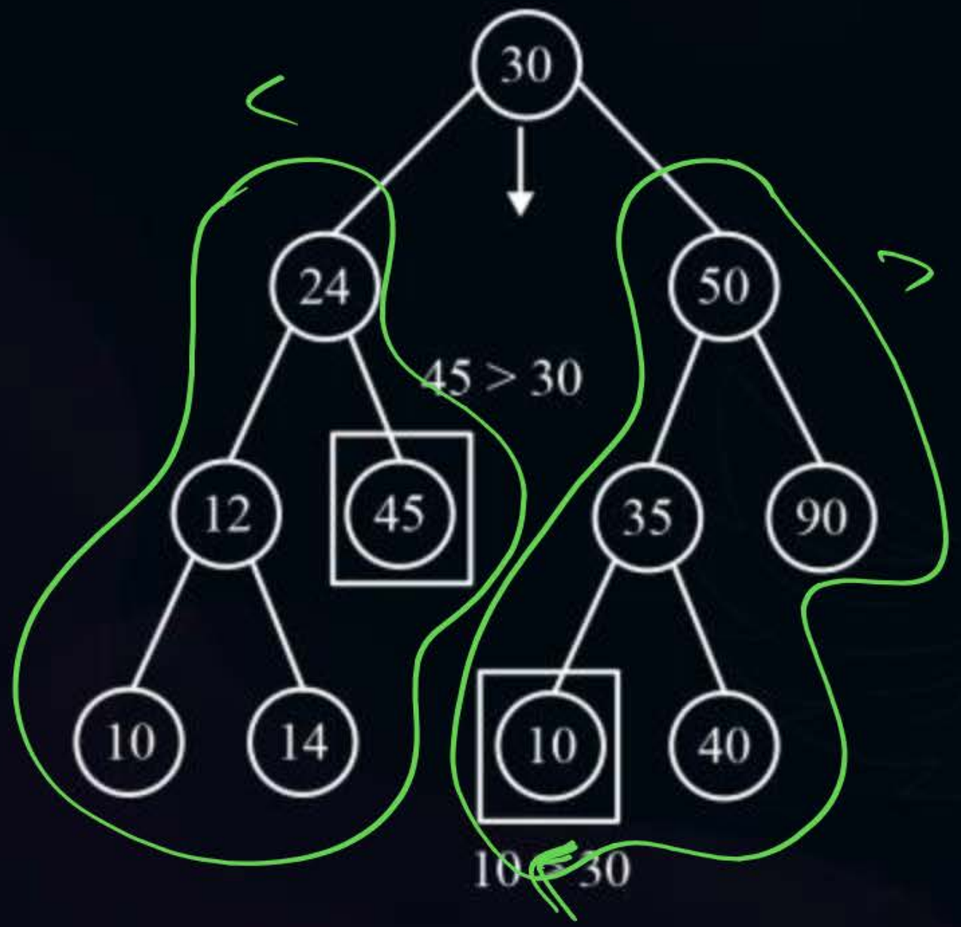


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~~Equation:~~ Eg:-

BST?

No



[NAT]

#Q. Is this a valid Binary tree?

Yes

[NAT]

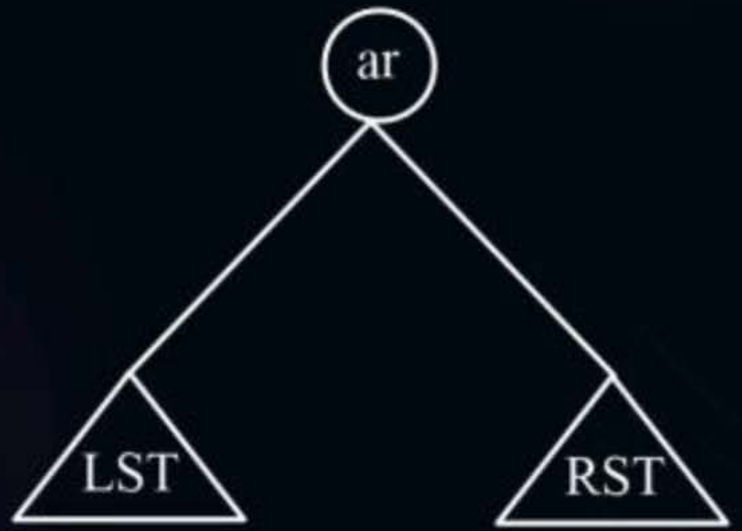
#Q. Is this a valid Binary Search tree? (BST)

No



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BST → General representation



All values in LST < ar < All values in RST



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Optimal cost binary Search tree (OBST)

[NAT]

#Q. For given data, which of the following Binary Search Trees (BSTs) will result in minimum cost? (Optimization Problem)

[NAT]

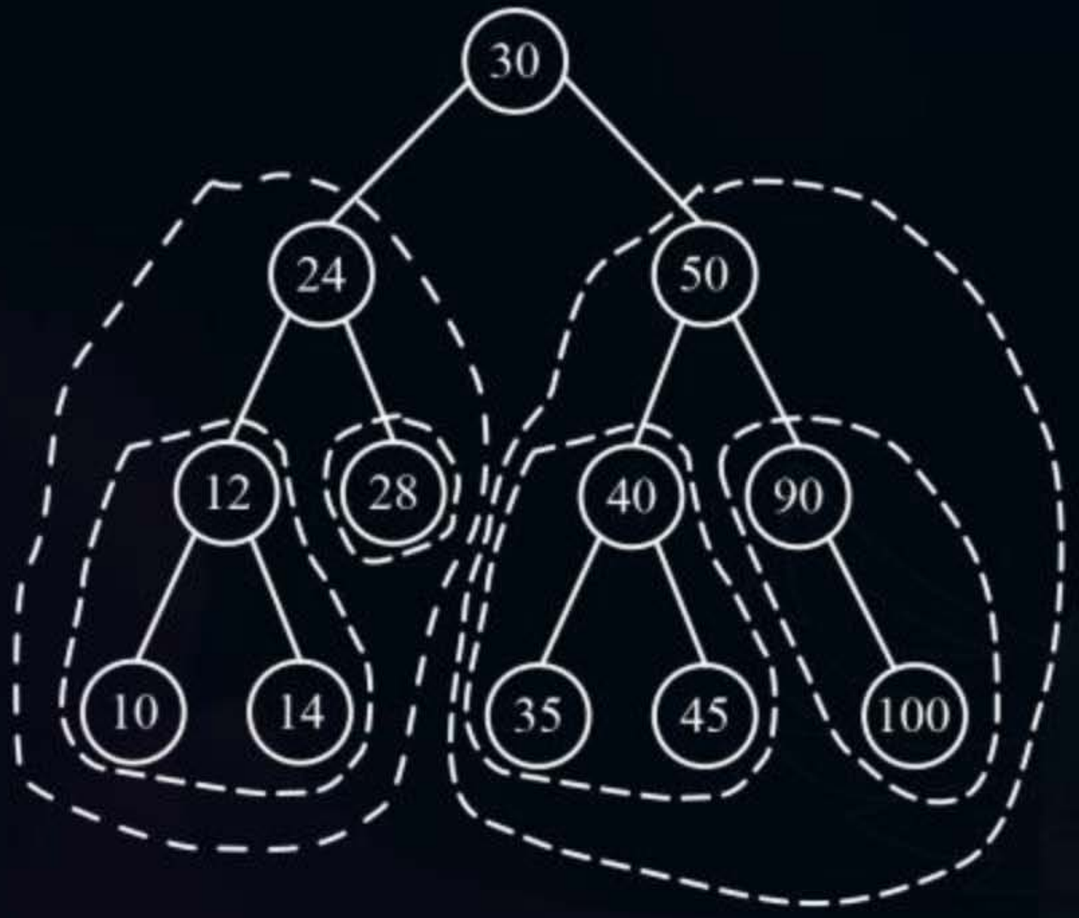
#Q. Construct the Binary Search Tree with minimum cost → DP based recurrence



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Equation:

Eg :-



Valid BST

valid ✓



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A Binary Search Tree for words / Text / Alphabets
(lots of applications in Computer Science)

Idea: Here comparison is made on the basis of lexicographical order.→ (dictionary order)

alphabetical (ASCII)

10 < 15

a b



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Equation:

a < b

ab < ad

a = a

b < d ?

adit > aditya → False

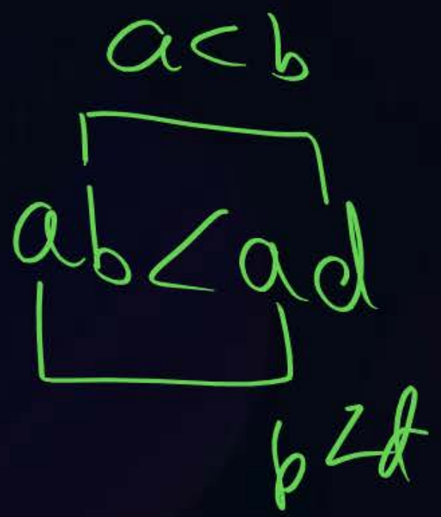
a = a

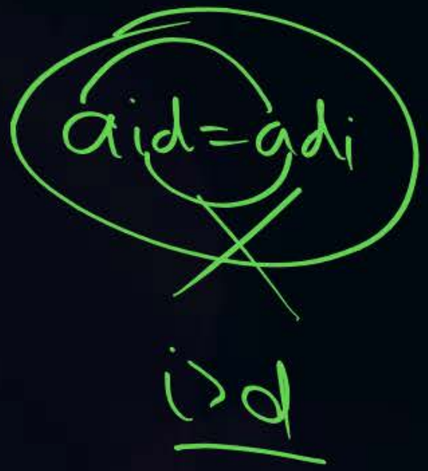
ad = ad

adi = adi

adit = adit

adit < aditya







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~~Equation-3~~ eg 3

Zebra < Aditya jain → False

z > a

abd > acz → False

b > c (✗)

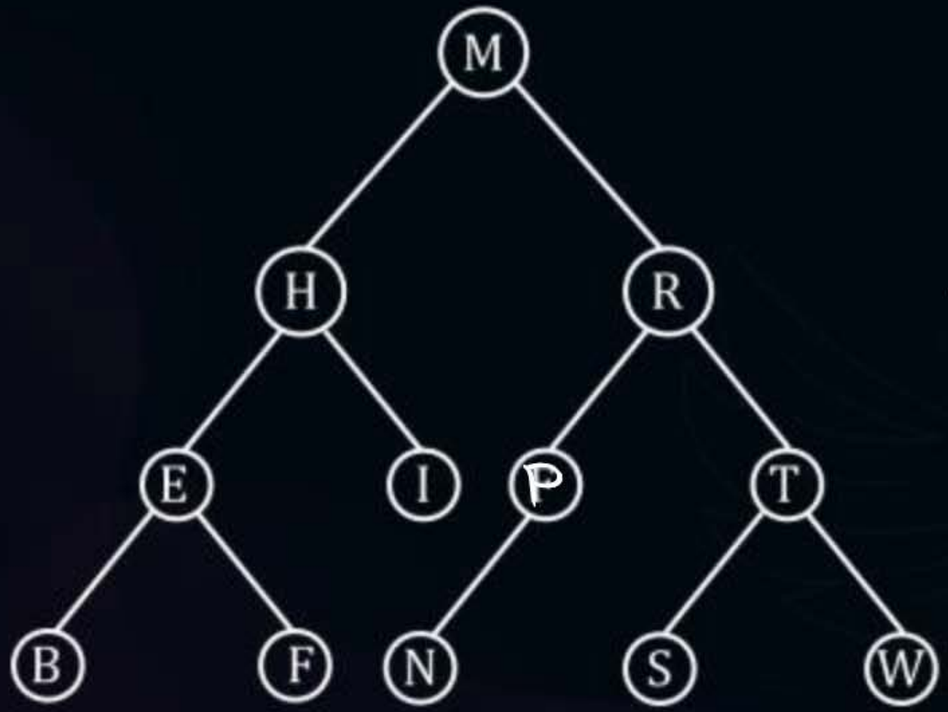


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Equation-2:

Is this a valid BST?

Yes





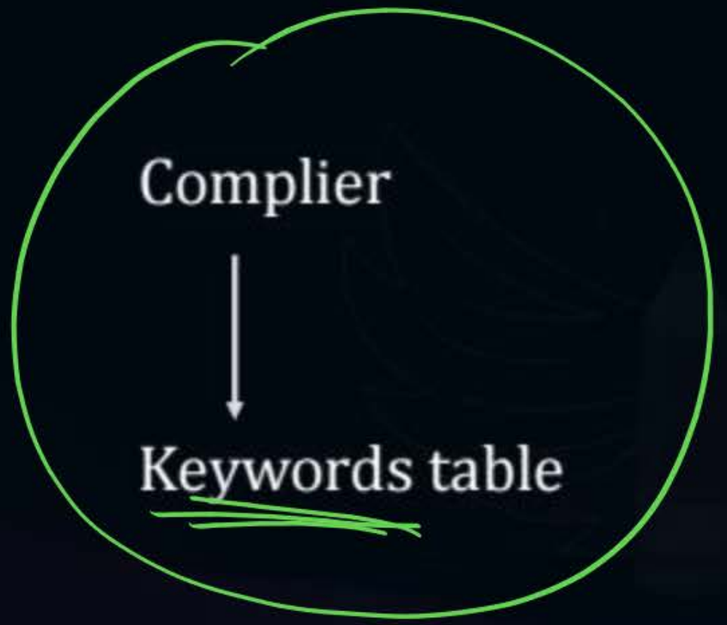
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Application of BSTs (in computer Science)

1. Efficient Searching
2. File/ data organization
3. Indexing

Keywords:

For, while, if , else





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In context of complies /programming language:

Keywords	Non-keywords
if, else	i, a,b aj,
for, while	first name
break	age
continue	
int	
return	

[NAT]

#Q. For n distinct elements, how many different binary search tree are possible?



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Catalan number:

Number of BSTs = $\frac{1}{(n+1)} *^{2n} C_n$



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Equation:

a, b, c

$$n = 3$$

$$n\frac{1}{(n+)} *^{2n} C_n$$

$$\frac{1}{(3+1)} *^{6} C_3$$

$$= \frac{1}{4} \times \frac{6!}{3! \times 3!}$$

$$= \frac{1}{4} \times \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} = 5$$



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Topic: Miscellaneous

Example:

A= [a_1, a_2, a_3] = [do, if , while]

do < if < while

Number of difference BSTs?

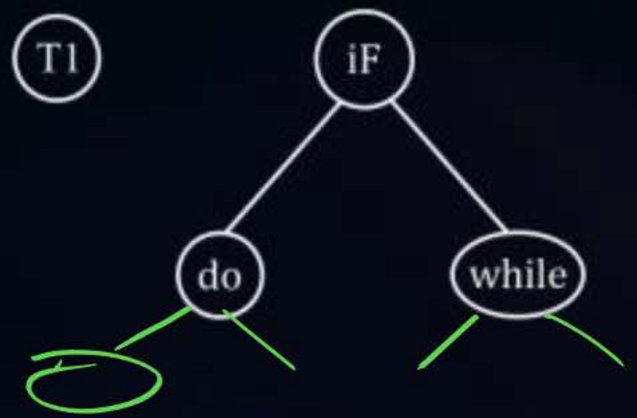
Number of identifiers n = 3

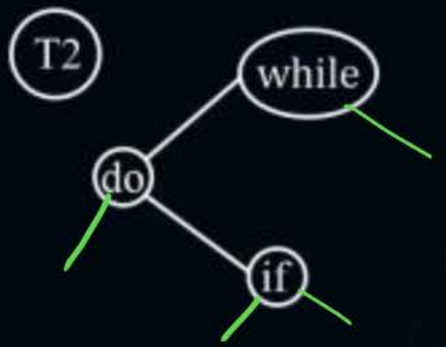
$$\frac{1}{(n+1)} \times {}^{2n}C_n = 5$$

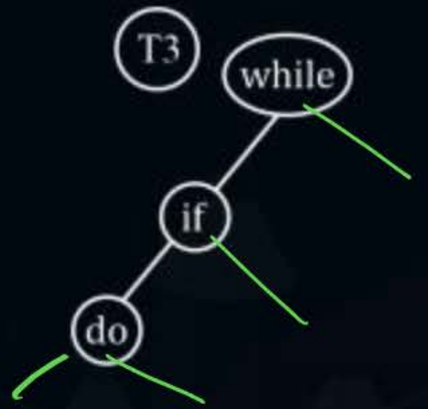
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Basic ~~infection~~ intuition

Prefer T1 as if has lower depth (min cost)







9









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Process to determine the cost of BST (T):

1. Cost (T) = cost (successful search) + cost (un-successful search)

Keyword

Non- Keyword

- A successful search will always end at a given tree node (non-null link)
- An un-successful search will always end at a null link.



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Note: Throughout the discussion, the assumption is that root node is at level. 1



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N → identifiers $(a_1, a_2 \ldots a_n)$

1. Cost (successful search) $= \text{cost}(a_1) + \text{cost}(a_2) \ldots \text{Cost}(a_n)$

$$= \sum_{i=1}^{n} \text{cost}(a_i)$$

2. Cost $(a_i) \propto$ level (a_i)

   Cost (a_i) = Pi * level (a_i)

   Where,

   Pi = the probability of the usefulness of this identifier a_i in the application.



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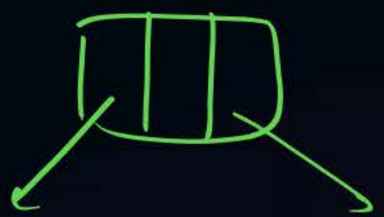
Example:

~~if, confines~~ if Continue

Usually, in programming ‘if’ is used more frequently then ‘continue’.

[NAT]

#Q. In a binary tree with n nodes, how many null links are there?





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Solution:

every node ⇒ ~~2 links~~ 2 links

Total ~~Tola~~ links = 2*n

~~Three~~ Tree with n nodes, has (n–1) non- null links

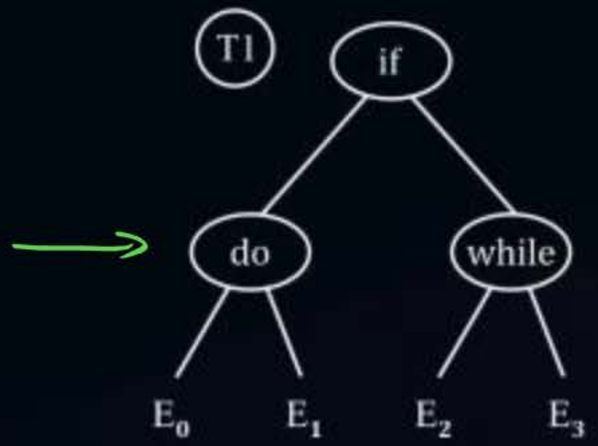
Hence, null links = Total– (non-null links)

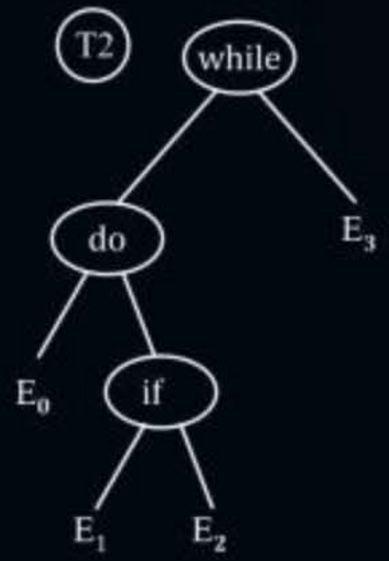
= 2n – (n–1)

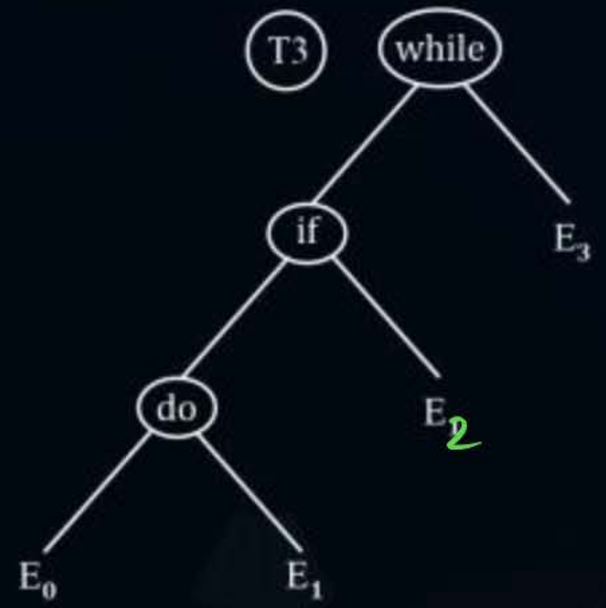
= n +1

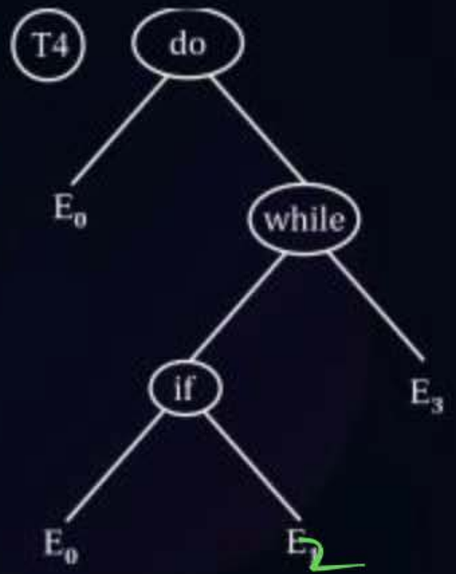


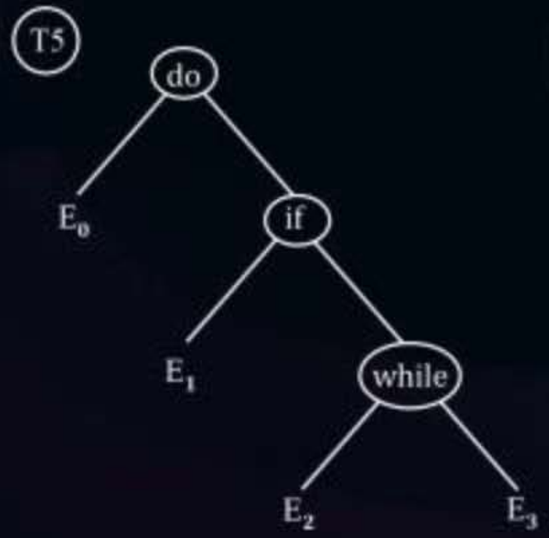
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N nodes → (n+1) null links $(E_0 \rightarrow E_n)$

1. Cost (unsuccessful search) = cost (E_0) +cos (E_1) .....+ cost (E_n)

$$= \sum_{i=0}^{n} cost(E_i)$$

2. Cost (E_i) ∝ (Level (E_i) –1)

   cost (E_i) = qi* (level (E_i)–1)

   qi → probability of Ei (non-keyword)



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Example:

Write a programme to add 2 integers.

1. Programme /Books / Tutorials → general pattern.

   int a, b, c, d

2. Define a for loop

   For (i = 1, i< =n ; i ++)

   For (aj =1 ; aj < n; aj ++)

   Pr (i) > Pr (aj)



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Summary: Cost of BST (T)

Cost (T) = Cost (successful search) + cost (un-successful search)

$$\text{Cost (T)} = \sum_{i=1}^{n} pi * level(ai) + \sum_{i=0}^{n} q_i * (level(Ei) - 1)$$

for solving questions:

Given:

$q_1 \rightarrow n$

$P_1 \rightarrow n$

$q_0 \rightarrow n$



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Given:

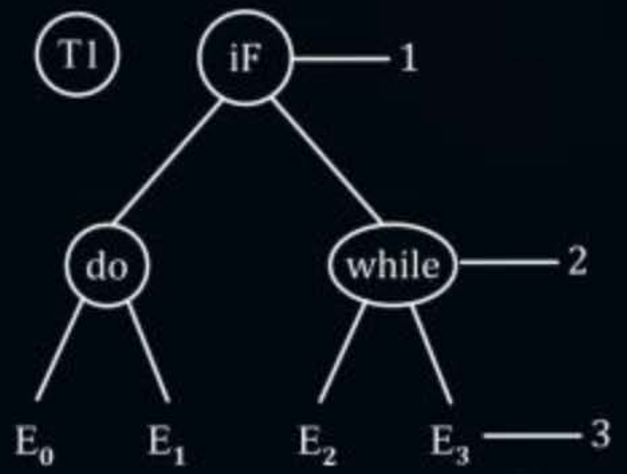
$A = [a_1, a_2, a_3] = [do, if, while]$

$[p_1, p_2, p_3] = [0.5, 0.1, 0.05]$

$[q_0, q_1, q_2, q_3] = [0.15, 0.1, 0.05, 0.05]$

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do → 0.5
if → 0.1
while → 0.05



$= (0.5 \times 2 + 0.1 \times 1 + 0.5 \times 2) + (0.15 \times (3-1) + 0.1 \times (3-1) + 0.05\,(3-1) + 0.05\,(3-1))$

$= 1 + 0.1 + 0.1 + 0.3 + 0.2 + 0.1 + 0.1$

$= 1.2 + 0.7 = 1.9$

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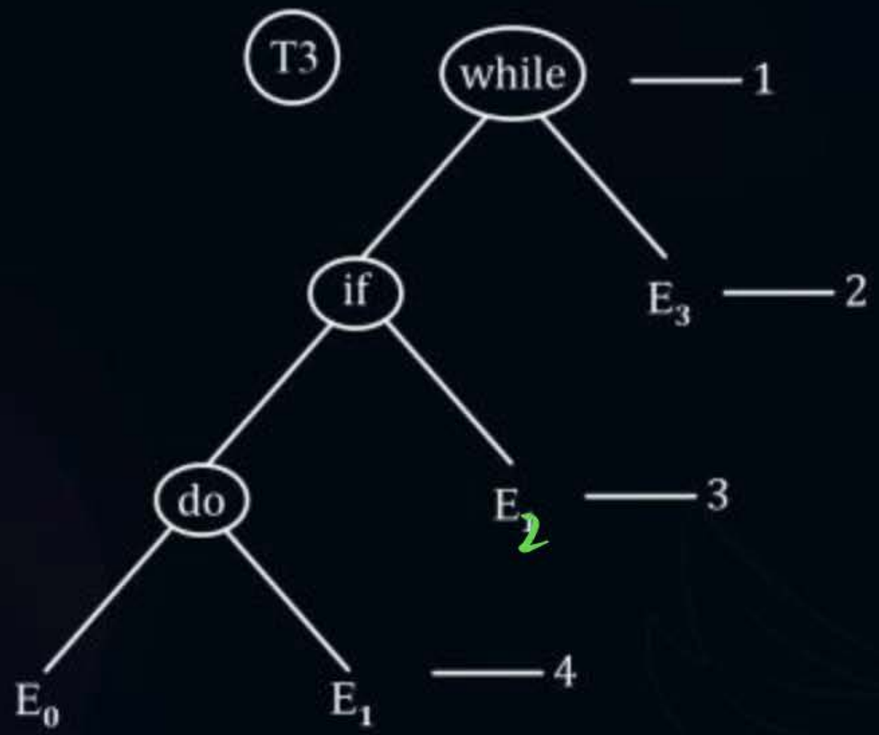
$= 0.5 \times 2 + 0.1 \times 3 + 0.05 \times 1 + 0.15 \times (3-1) + 0.1 \times (4-1) + 0.05\,(4-1) + 0.05(2-1)$

$= 1 + 0.3 + 0.05 + 0.3 + 0.3 + 0.15 + 0.05$

$= 2.15$



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$0.5 \times 3 + 0.1 \times 2 + 0.05 \times 1 + 0.15 \times (4-1) + 0.1 \times (4-1) + 0.05 \times (3-1) + 0.05 \times (2-1)$

$= (1.5 + 0.2 + 0.05) + (0.45 + 0.3 + 0.1 + 0.05)$

$= 2.65$

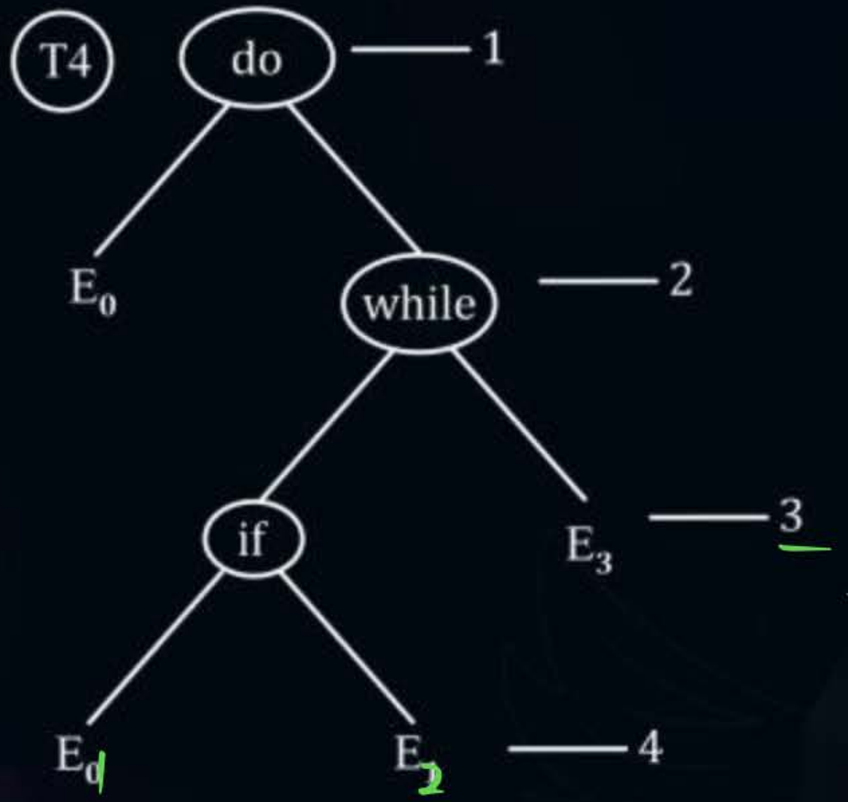
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given

do → 0.5

if → 0.1

while → 0.05

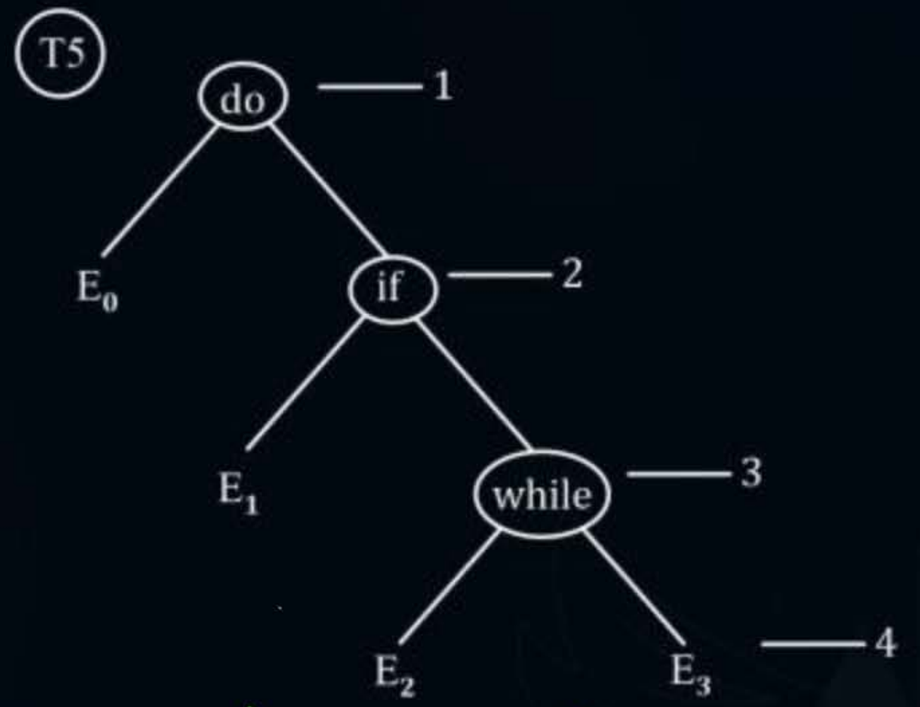


Cost (T4)= ?

= (0.5 × 1 + 0.1×3 + 0.05 ×2) + (0.15 × (2–1) + 0.1 × (4–1) + 0.05 (4–1) + 0.05 (3–1)

= 1.6

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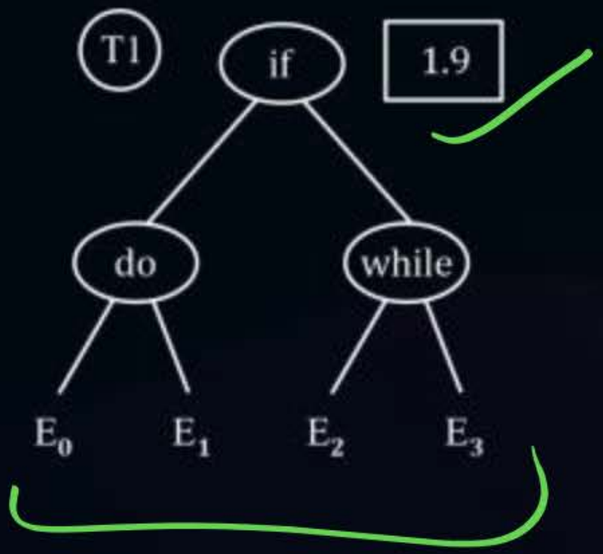
Cost (T5) = ?

$= (0.5 \times 1 + 0.1 \times 2 + 0.05 \times 3) + (0.15 \times (2-1) + 0.1 \times (3-1) + 0.05 \times (4-1) + 0.05 \times (4-1)$

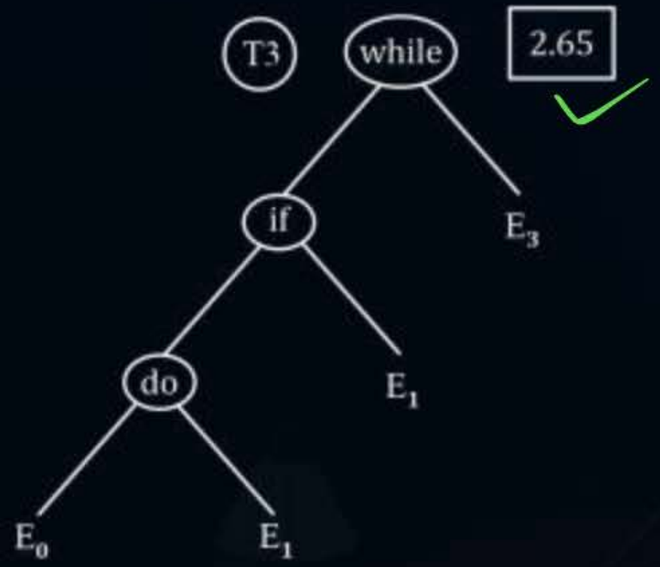
$= (0.5 + 0.2 + 0.15) + (0.15 + 0.2 + 0.15 + 0.15)$

$= 1.5$

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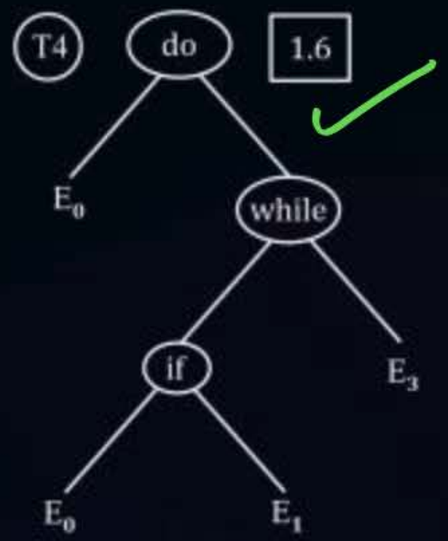


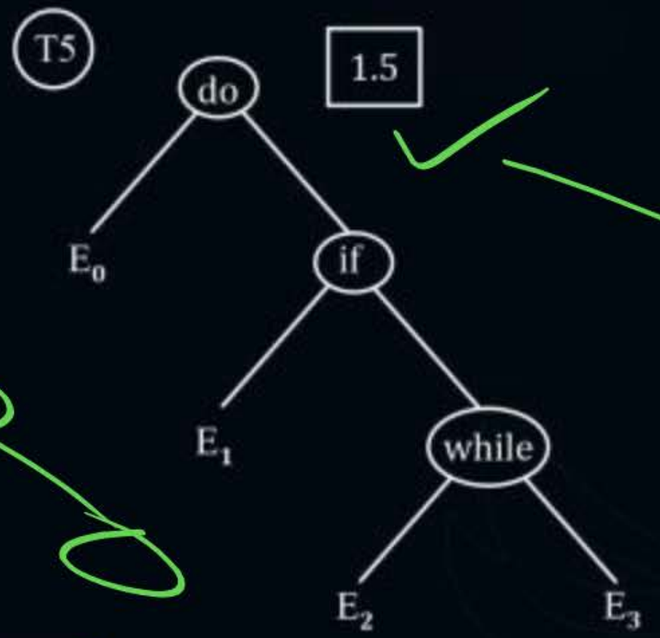






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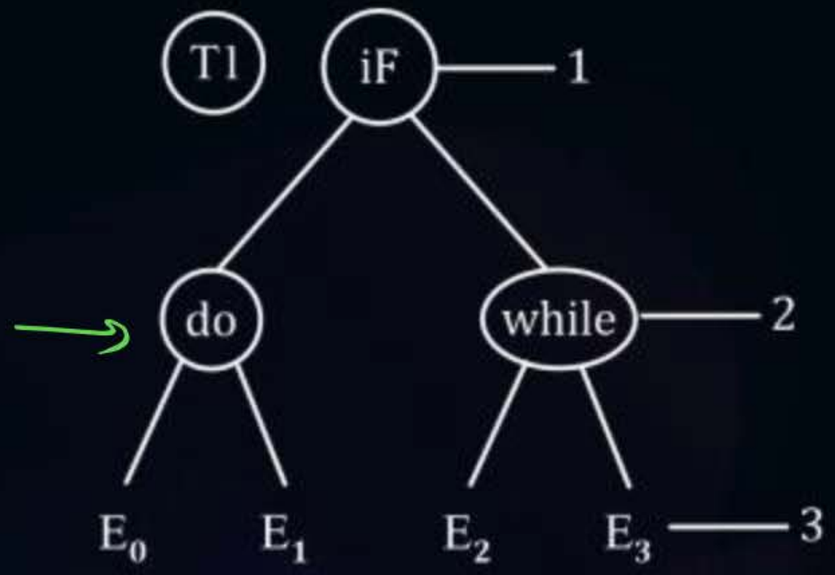


actual min.

(OBST)

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Observation:



Cost = 1.9

do → 0.5

If → 0.1

While→ 0.5



Cost = 1.5



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Imp:-

The cost of the BST depends on both:

The depth (levels)as well as the probabilities of all the identifiers(pi & qi)(p_i & q_i)

Idea is very similar to Huffman Encoding / Optimal Merge Pattern.



2 mins Summary

→ OBST



THANK - YOU